An Adaptive Physics-Informed Framework for Subsea Pipeline Reliability and Maintenance

Devesh Shukla, and Vinay Kumar

***Abstract*—Subsea pipelines are exposed to progressive corrosion and discrete external or operational shocks whose interaction introduces substantial uncertainty into reliability assessment, remaining useful life (RUL) prediction, and maintenance planning. Existing stochastic degradation, competing-failure, and Dynamic Bayesian Network (DBN) approaches address different aspects of this problem, but degradation modeling, sequential parameter adaptation, probabilistic prognosis, and maintenance decision-making are often treated separately. This paper develops an Adaptive Physics-Informed Dynamic Bayesian Network (API-DBN) that integrates these functions within a unified condition-updating framework. Stochastic corrosion is represented by a Wiener process and shock occurrence by a Poisson process, with a temporally ordered, state-dependent coupling mechanism describing their interaction. Sequential Monte Carlo filtering with Markov Chain Monte Carlo rejuvenation jointly updates the latent degradation state and selected evolving model parameters as inspection observations become available. The resulting posterior is propagated through Monte Carlo simulation to obtain probabilistic RUL estimates, which are subsequently incorporated into a risk–cost maintenance model. In controlled simulations of 9,990 pipelines whose degradation parameters differ between pipelines and drift over time, API-DBN reduces the RUL mean absolute error by 19% and the continuous ranked probability score by 29% relative to a static-parameter DBN, and its 95% prediction intervals contain the true RUL in 95.3% of forecasts, against 59.9% for the static DBN. When each model drives the inspection and replacement decisions for the same simulated pipelines, the API-DBN-based policy lowers the expected discounted lifecycle cost by 17% relative to the static-DBN-based policy. The benefit of adaptation arises when degradation parameters vary across assets or over time; with identical, known parameters a static DBN performs better. Public BSEE/NETL Gulf of Mexico pipeline and incident records are used to check the geometry, shock-rate, and failure-cost assumptions.**

***Index Terms*—Subsea Pipelines, Dynamic Bayesian Networks, Dependent Competing Failures, Remaining Useful Life, Sequential Monte Carlo, Maintenance Optimization.**

I. Introduction

Subsea pipelines are critical components of offshore energy infrastructure, transporting hydrocarbons between production facilities, processing systems, and onshore terminals. Their integrity is challenged by prolonged exposure to corrosive seawater, high hydrostatic pressure, operational loading, and accidental external actions such as dropped objects, anchor interaction, and trawling [1]. Historical Gulf of Mexico incident records further demonstrate that pipeline failures remain an important integrity-management concern [2], [3]. Because subsea inspection, repair, and replacement are costly and operationally constrained, reliable condition assessment and timely maintenance planning are essential for managing structural risk over the service life of these assets.

Subsea pipeline deterioration involves multiple mechanisms whose effects can coexist and interact. Corrosion produces progressive wall loss, whereas external or operational shocks can introduce discrete damage and modify subsequent deterioration. Corrosion-induced wall thinning can reduce structural resistance to later loading, while impact-induced damage to protective systems can accelerate subsequent corrosion. Microbiologically influenced corrosion [4] and fatigue are also important contributors to subsea-pipeline deterioration; however, the present study focuses specifically on the dependent interaction between progressive corrosion and discrete shock-induced damage. Treating these mechanisms as independent can therefore neglect dependencies that influence failure probability and remaining useful life (RUL).

Existing integrity-assessment methods address different aspects of this problem. Deterministic physics-based models provide interpretable relationships between deterioration and structural capacity but can be sensitive to parameters established during initial calibration. Stochastic degradation models, including Wiener and Gamma processes, represent variability in deterioration evolution [5], [6], while dependent competing-failure models provide mechanisms for representing interactions between degradation and shocks. Dynamic Bayesian Networks (DBNs) extend probabilistic reliability assessment across successive time slices and allow uncertain system states to be updated as evidence becomes available [7]. DBN applications to subsea pipelines have further demonstrated the ability to represent degradation, competing-failure processes, and parameter uncertainty [8], [9]. These developments provide important components for condition-based reliability assessment, but they do not by themselves establish a complete adaptive prognosis-to-decision framework.

The remaining challenge is primarily one of methodological integration. Degradation modeling, dependent-failure analysis, Bayesian state estimation, parameter calibration, RUL prediction, and maintenance optimization are often addressed as separate stages. Consequently, new inspection evidence may update an estimate of the current pipeline condition without consistently revising uncertain degradation parameters, propagating the resulting posterior uncertainty to future failure time, and transferring that information to the maintenance-decision layer. In addition, existing competing-failure formulations represent several forms of degradation–shock dependence, includ-

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ing dependence in both directions [10], but such coupling has rarely been embedded in a sequentially updated DBN whose parameters adapt to inspection data. A condition-updating framework is therefore needed in which inspection evidence jointly informs the evolving degradation state and selected model parameters, dependent corrosion–shock behavior is represented explicitly, and the updated posterior is propagated from reliability assessment to probabilistic RUL prediction and maintenance planning. Section II examines these limitations in detail and establishes the corresponding research gaps.

To address this problem, this study develops an Adaptive Physics-Informed Dynamic Bayesian Network (API-DBN) for subsea pipelines subjected to interacting progressive corrosion and discrete shocks. Here, "physics-informed" refers to embedding stochastic corrosion kinetics, shock-arrival behavior, state-dependent corrosion–shock interaction, and a structural failure criterion within the probabilistic state-transition model. Continuous corrosion is represented by a Wiener process and shock occurrence by a Poisson process, with their interaction introduced through a temporally ordered state-dependent coupling mechanism. The resulting physical states and uncertain baseline parameters are embedded within a time-sliced DBN. As inspection observations become available, Sequential Monte Carlo (SMC) filtering with Metropolis–Hastings MCMC rejuvenation updates the posterior distribution of the latent degradation state and selected degradation and shock parameters. This posterior is subsequently propagated forward by Monte Carlo simulation to obtain a probabilistic RUL distribution, which is incorporated into a risk–cost model for inspection and preventive-maintenance planning.

The principal contributions of this study are as follows:

1) A physics-informed dependent degradation formulation with a continuous parametric corrosion–shock coupling: a sigmoid acceleration of the corrosion drift driven by the total equivalent damage, and an exponential dependence of the shock intensity on the corrosion depth, both acting with temporal ordering across DBN time slices.
2) An adaptive DBN formulation with SMC–MCMC inference that sequentially updates the latent degradation state and selected evolving baseline parameters as new inspection information becomes available, rather than relying exclusively on a single offline parameter calibration.
3) An integrated prognosis-to-decision procedure in which the updated posterior distribution is propagated to probabilistic RUL and subsequently incorporated into a risk–cost maintenance model for selecting inspection intervals and preventive-maintenance thresholds.
4) A controlled simulation study with known failure times in which pipelines differ in their degradation parameters, comprising matched baselines, a factorial ablation of coupling and MCMC rejuvenation, an oracle reference, and a closed-loop evaluation of maintenance policies against the simulated truth.

The experimental design deliberately distinguishes controlled predictive validation from historical application. Quantitative RUL accuracy is evaluated using controlled simulations for which failure times are known, whereas the BSEE/NETL records are used to check the parameter assumptions. The historical records do not provide complete longitudinal inspection and true-RUL histories for pipeline segments; consequently, the simulation-based RUL results are not interpreted as direct historical RUL validation.

The remainder of this paper is organized as follows. Section II reviews related work and establishes the research gap. Section III presents the problem formulation and proposed API-DBN methodology, including the dependent degradation model, sequential Bayesian inference, probabilistic RUL prediction, and maintenance-decision formulation. Section IV describes the historical data, controlled simulation design, baselines, and evaluation protocol. Section V presents and discusses the predictive, maintenance, and computational results. Finally, Section VI summarizes the principal findings, limitations, and directions for future work.

II. Related Work and Research Gap

Research on subsea-pipeline reliability spans stochastic degradation modeling, competing-failure analysis, probabilistic graphical models, remaining useful life (RUL) prediction, and maintenance optimization. These research streams address complementary aspects of integrity management, but their integration is particularly important when progressive deterioration and discrete damage events interact and inspection information becomes available sequentially. This section reviews the literature from that perspective and identifies the specific methodological gaps addressed by the proposed API-DBN.

A. Stochastic Degradation and Dependent Competing-Failure Models

Progressive corrosion is commonly represented using stochastic degradation processes that account for uncertainty in the evolution of material loss. Wiener-process formulations are attractive because of their analytical tractability and ability to represent stochastic variability in degradation increments [5]. Gamma-process models provide an alternative when monotonic degradation is required [6]. These formulations establish a probabilistic basis for estimating deterioration and structural reliability, although their predictive performance depends on how adequately the assumed process and its parameters represent changing operating and environmental conditions.

Pipeline reliability is further complicated by the simultaneous presence of progressive degradation and discrete shocks. Wang et al. [11] considered reliability assessment under degradation and shock processes, while Peng et al. [12] developed reliability and maintenance models for systems subject to multiple dependent competing-failure processes. Such formulations demonstrate that deterioration and shock mechanisms cannot always be treated as isolated contributors to system failure.

Subsequent studies have examined more explicit forms of dependence between the competing mechanisms. Rafiee et al. [13] modeled dependent competing-failure processes in which shock events can change the degradation rate, and later

generalized the treatment of competing risks through mixed shock models [14]. Sun et al. [10] further modeled dependence in both directions, considering the effect of random shocks on the increments and rate of a nonlinear Wiener degradation process together with the effect of degradation on the shock process, and used a time-varying copula to describe the dependence among failure processes. These approaches provide an important basis for representing interactions between continuous deterioration and discrete damage events.

For subsea pipelines the interaction is likewise reciprocal: accumulated external damage may alter the subsequent corrosion environment or effective corrosion rate, while progressive wall loss reduces structural resistance and modifies the effective susceptibility to later shocks. The present study therefore does not claim to introduce two-way degradation–shock dependence. Its distinguishing elements are (i) a specific continuous parametric form for each direction, namely a sigmoid acceleration of the corrosion drift driven by the total equivalent damage and an exponential dependence of the shock intensity on the corrosion depth (Section III-A), and (ii) the embedding of this coupling in a time-sliced DBN whose latent state and baseline parameters are updated sequentially from inspection data (Section III-B).

B. Dynamic Bayesian Reliability and Sequential Updating

Bayesian networks provide a probabilistic representation of conditional dependencies among uncertain variables, while Dynamic Bayesian Networks (DBNs) extend this representation across successive time slices. Straub [7] demonstrated the use of DBNs for stochastic deterioration processes, providing a general framework for representing temporal dependence and updating uncertain system states as evidence becomes available.

DBN formulations have subsequently been applied to offshore and subsea reliability problems. Cai et al. [15] developed a DBN-based resilience-assessment approach for subsea oil and gas pipelines. More directly related to the present problem, Liu et al. [8] considered dependent competing-failure processes of subsea pipelines under parameter uncertainty using a DBN. These studies demonstrate the suitability of temporal probabilistic graphical models for integrating degradation mechanisms, uncertain variables, and reliability states.

An important distinction, however, exists between updating the probability distribution of a latent degradation state and allowing the parameters governing the degradation process to evolve as new evidence is assimilated. A model may be dynamic at the state level while retaining parameters obtained through offline calibration. When corrosion rate, shock frequency, or degradation variability changes over time, updating the state alone may therefore be insufficient to represent the evolving condition of the asset.

Sequential Monte Carlo methods provide a natural mechanism for approximating posterior distributions in nonlinear and non-Gaussian state-space models. Particle filtering propagates a population of weighted samples as observations arrive, and Markov chain Monte Carlo moves applied after resampling (the resample–move algorithm) restore particle diversity [16]. Particle methods for joint inference over latent states and uncertain parameters are reviewed in [17].

The relevant gap is therefore narrower than the absence of dynamic Bayesian reliability models. DBNs for deterioration and subsea-pipeline reliability already exist. What remains limited in the literature reviewed here is the integration of a physics-informed DBN with sequential joint updating of the latent degradation state and selected time-evolving baseline degradation parameters as new inspection observations become available. This distinction motivates the adaptive DBN and SMC–MCMC formulation presented in Section III-B.

C. RUL Prediction and Maintenance Decision Support

Reliability assessment and RUL prediction address related but distinct questions. Reliability characterizes the probability that a system continues to satisfy its performance requirements over a specified period, whereas RUL prediction concerns the distribution of the time remaining before a defined failure condition is reached. For degrading infrastructure, the latter is particularly useful when inspection and maintenance decisions must respond to the current estimated condition.

Probabilistic degradation models can be propagated forward to obtain failure-time or RUL distributions, and competing-failure formulations have also been linked to maintenance analysis. Peng et al. [12], for example, explicitly considered both reliability and maintenance for systems subject to dependent competing failure processes. Risk-based inspection methods have likewise been developed for corroded subsea pipelines [18], demonstrating the importance of connecting deterioration uncertainty with inspection planning.

The principal issue for the present study is therefore not the absence of maintenance optimization or RUL methods. Rather, degradation estimation, probabilistic prognosis, and maintenance optimization are often treated as distinct stages with different assumptions and information structures. In such a setting, newly acquired inspection evidence may update an estimate of current condition without being propagated consistently through uncertain model parameters, future failure-time distributions, and the resulting maintenance decision.

This distinction becomes important when maintenance policies are defined by fixed inspection intervals or deterministic intervention thresholds. Such policies may not fully exploit changes in the posterior degradation state or uncertainty revealed by subsequent inspections. Propagating the updated posterior distribution to RUL and then incorporating RUL uncertainty into a risk–cost formulation provides a direct mechanism for linking condition information with inspection and preventive-maintenance decisions.

Accordingly, the methodological requirement considered in this study is an inference-to-decision chain in which inspection observations update the degradation model, the updated posterior is propagated to a probabilistic failure-time distribution, and that distribution enters the maintenance objective. Section III-C develops this connection for inspection-interval and preventive-maintenance-threshold selection.

D. Research Gap and Positioning of the Present Study

Table I compares representative approaches according to the methodological capabilities most relevant to the present study. The comparison is intended to clarify how stochastic degradation, competing shocks, dependency modeling, dynamic probabilistic inference, online updating, RUL prediction, and maintenance optimization have been addressed across the literature. The individual techniques are not themselves absent from prior work; the principal issue is the extent to which they are integrated within a common condition-updating framework.

The literature synthesis and the comparison in Table I identify four closely related research gaps.

1) Limited joint state–parameter adaptation. Existing stochastic degradation and DBN approaches provide mechanisms for representing deterioration uncertainty and, in some cases, updating system states as new information becomes available. However, the literature reviewed here provides limited integration of online inspection evidence with joint sequential estimation of the latent degradation state and selected parameters governing corrosion, shock occurrence, and process uncertainty. This limitation is particularly relevant when the underlying degradation environment is nonstationary and parameters obtained from offline calibration may no longer adequately represent the current asset condition.

2) Limited integration of dependent corrosion–shock evolution within adaptive inference. Competing-failure models have established that progressive degradation and random shocks can be statistically dependent, and dependence in both directions has been modeled [10]. In the reviewed literature, however, such dependence is not expressed as a continuous parametric coupling, in which accumulated shock damage accelerates subsequent corrosion while the corrosion state modifies the effective shock intensity, embedded in a sequentially updated DBN. A formulation of this type must also avoid introducing a directed cycle within a DBN time slice.

3) Fragmentation between degradation inference and probabilistic prognosis. Estimating the current degradation state or reliability does not by itself provide the posterior distribution of the time remaining until structural failure. For maintenance planning under uncertainty, the updated state and parameter distributions must be propagated through the physical failure model to obtain a probabilistic RUL distribution. The literature reviewed here addresses degradation modeling, Bayesian reliability assessment, and RUL prediction to different extents, but their integration within the same adaptive condition-updating chain remains limited.

4) Limited integration of updated prognosis with maintenance decisions. Maintenance and risk-based inspection optimization are established research areas, including formulations for systems subject to dependent competing failures. The unresolved integration issue is whether a new inspection observation can propagate consistently from posterior state and parameter updating to RUL uncertainty and subsequently to a risk–cost maintenance policy. Without this connection, maintenance decisions may remain based on fixed schedules, offline parameter estimates, or prognostic quantities that do not fully reflect the most recent condition information.

These gaps motivate the API-DBN developed in Section III. The proposed framework combines a physics-informed stochastic degradation model with a temporally ordered corrosion–shock interaction, sequential SMC–MCMC inference over latent states and selected evolving baseline parameters, posterior Monte Carlo RUL prediction, and risk–cost maintenance optimization. The intended novelty therefore lies primarily in the integration and adaptive coupling of these elements rather than in claiming that Wiener degradation, Poisson shocks, DBNs, particle filtering, MCMC, RUL prediction, or maintenance optimization are individually new.

More specifically, the four gaps established above map directly to the methodological development in Section III: the degradation–failure and corrosion–shock interaction model addresses the dependent-degradation representation in Section III-A; the adaptive DBN and SMC–MCMC procedure address joint state–parameter updating in Section III-B; and posterior forward propagation together with the lifecycle-cost model connect probabilistic RUL prediction to maintenance decisions in Section III-C. This correspondence provides the basis for the experimental evaluation in Sections IV and V.

III. Problem Formulation and Proposed API-DBN Methodology

The Adaptive Physics-Informed Dynamic Bayesian Network (API-DBN) integrates stochastic degradation modeling, sequential Bayesian inference, probabilistic remaining useful life (RUL) prediction, and maintenance optimization within a unified reliability framework for subsea pipelines. The modeled degradation state comprises continuous corrosion damage and cumulative equivalent damage induced by discrete shocks. Inspection observations are assimilated sequentially to update both the latent degradation state and selected baseline model parameters. The resulting posterior distribution is then propagated forward to characterize uncertainty in future failure time and to support risk–cost-based inspection and preventive-maintenance decisions.

Figure 1 summarizes the complete information flow. Historical pipeline and incident records are first used to check the parameter assumptions of the degradation and cost models. Continuous corrosion and discrete shock processes are then embedded within a time-sliced DBN, where new inspection observations trigger SMC–MCMC updating of the latent states and adaptive baseline parameters. The updated posterior is propagated forward through Monte Carlo simulation to obtain a probabilistic RUL distribution, which subsequently enters the maintenance-optimization model. Quantitative RUL performance is evaluated using controlled simulations with known failure times.

A. System Definition and Degradation–Failure Model

At inspection epoch k, the latent physical state is represented by $\mathbf{s}_k = (X_k, S_k)$, where X_k denotes corrosion depth and S_k denotes cumulative equivalent shock damage. The corresponding total equivalent damage is $Z_k = X_k + S_k$. The adaptive baseline parameter vector is $\theta_k = (\mu_k^{(0)}, \lambda_k^{(0)}, \sigma_{B,k})$,

TABLE I
METHODOLOGICAL COMPARISON OF REPRESENTATIVE APPROACHES RELEVANT TO DEPENDENT DEGRADATION, DYNAMIC RELIABILITY ASSESSMENT, AND MAINTENANCE PLANNING

Study	Degradation Model	Shock / Competing Failure	Dependency	Dynamic Model	Online Updating	RUL	Maintenance
Wang et al. [11]	Wiener	Poisson shocks	Shock→degradation (level or rate)	No	No	No	No
Peng et al. [12]	Gamma	Poisson shocks	Shock→degradation (abrupt damage)	No	No	No	Yes
Rafiee et al. [13]	Gamma	Random shocks	Rate-dependent	No	No	No	No
Straub [7]	General deterioration	No	–	DBN	Yes (inspection data)	No	No
Liu et al. [8]	Wiener	Poisson shocks	Shock→degradation	DBN	Parametric (offline)	No	No
Cai et al. [15]	General	No	No	DBN	No	No	No
Sun et al. [10]	Nonlinear Wiener	Poisson shocks	Two-way (copula-based)	No	No	No	No
Proposed API-DBN	**Wiener**	**Poisson shocks**	**Two-way; continuous parametric**	**DBN**	**SMC–MCMC (state and parameters)**	**Probabilistic**	**Risk–cost optimization**

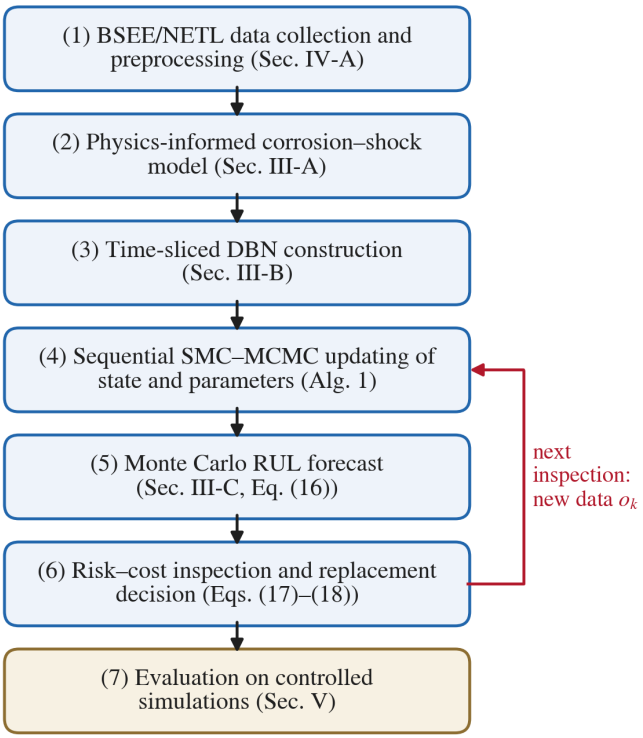


Fig. 1. End-to-end API-DBN workflow. The degradation and failure model is defined in Section III-A, sequential DBN inference is described in Section III-B, and posterior RUL prediction and maintenance optimization are formulated in Section III-C. The feedback arrow indicates that the maintenance decision schedules the next inspection, whose observation re-enters the DBN and triggers an SMC–MCMC update.

where $\mu_k^{(0)}$ is the baseline corrosion drift, $\lambda_k^{(0)}$ is the baseline shock intensity, and $\sigma_{B,k}$ represents degradation-process uncertainty. State-dependent effective quantities μ_k^{eff} and λ_k^{eff} govern the physical propagation of corrosion and shock occurrence, respectively.

Table II summarizes the principal variables used in the API-DBN formulation.

TABLE II
DBN NODES AND DESCRIPTIONS

Node	Description
X_k	Corrosion depth (continuous, latent)
S_k	Cumulative equivalent shock damage (continuous, latent)
$\mu_k^{(0)}, \mu_k^{\text{eff}}$	Baseline / effective corrosion drift
$\lambda_k^{(0)}, \lambda_k^{\text{eff}}$	Baseline / effective shock intensity
$\sigma_{B,k}$	Diffusion coefficient (time-varying)
Z_k	Equivalent total damage ($X_k + S_k$)
F_k	Binary failure indicator (Eq. (9))
O_k	Observation node (inspection measurement)

Continuous corrosion process. Wiener processes are widely used for corrosion-depth modeling because their normally distributed increments provide a tractable representation of stochastic degradation variability [5]. Let $X(t)$ denote the cumulative wall-thinning depth due to corrosion at time t. The corrosion process is modeled as

$$dX(t) = \mu^{\text{eff}}(t,\theta)\,dt + \sigma_B\,dB(t), \tag{1}$$

where $\mu^{\text{eff}}(t,\theta)$ is the time-varying effective corrosion drift, σ_B is the diffusion coefficient, and $B(t)$ is standard Brownian motion.

Unlike a conventional constant-drift formulation [5], the effective drift is allowed to vary through both the slowly evolving baseline parameter introduced later and the state-dependent corrosion–shock coupling. Although a Gamma process guarantees monotonic degradation [6], the Wiener process is retained here to represent stochastic variability while maintaining analytical tractability for sequential filtering.

For a time increment Δt, the corresponding transition density is

$$f_X(x_k \mid x_{k-1}) = \frac{1}{\sqrt{2\pi\sigma_B^2\Delta t}} \exp\left(-\frac{(x_k - x_{k-1} - \mu_k^{\text{eff}}\Delta t)^2}{2\sigma_B^2\Delta t}\right). \quad (2)$$

Equation (2) assumes a constant effective drift over each interval Δt, whereas the baseline drift $\mu^{(0)}$ may evolve across successive inspection epochs. Independent increments are assumed. This formulation can produce non-physical negative increments at the latent-state level, which remains a limitation relative to strictly monotonic Gamma-process formulations. Process uncertainty and inspection uncertainty are treated separately: σ_B governs physical state-process variability, whereas the observation-noise variance introduced in Section III-B represents measurement uncertainty.

Discrete shock process and equivalent damage. Operational and environmental impacts are represented as random shock events using a time-varying Poisson process [14]. For the interval $(t_{k-1}, t_k]$, the number of shock arrivals satisfies

$$N_k - N_{k-1} \sim \text{Poisson}\left(\int_{t_{k-1}}^{t_k} \lambda^{\text{eff}}(u)\, du\right). \quad (3)$$

The Poisson formulation provides a tractable representation of random shock occurrence while allowing the effective arrival intensity λ^{eff} to vary with the degradation state. Arrivals are therefore conditionally independent given the intensity path $\{\lambda^{\text{eff}}(u)\}$, but not unconditionally independent, because λ^{eff} itself depends on the evolving degradation state (Eq. (6)). Shock magnitudes are independent and identically distributed and independent of the arrival times.

Each shock has a physical magnitude Y_i, such as dent depth or impact energy. In the spirit of the cumulative-damage shock models of [14], the shock is mapped to an equivalent wall-loss increment through the power law

$$\Delta S_i = \kappa\, Y_i^{\eta}, \quad (4)$$

where $Y_i \sim \mathcal{N}(\mu_{y_0}, \sigma_{y_0}^2)$, truncated at zero, and κ and η are material- and geometry-dependent coefficients. With the values in Table IV, the mean equivalent wall loss per shock is $\mathbb{E}[\Delta S_i] = \kappa\, \mathbb{E}[Y_i^{\eta}] \approx 0.17$ mm. The cumulative equivalent shock damage is

$$S(t) = \sum_{i:t_i \le t} \Delta S_i.$$

The power-law form of Eq. (4) is a modelling assumption of this study, and its coefficients require empirical calibration for a specific asset class, which is an important limitation of this representation.

State-dependent corrosion–shock coupling. Existing step-change formulations can represent a shock-induced change in degradation rate through abrupt transitions [10], [13]. The present framework instead uses a continuous sigmoid relationship. Let $\mu_k^{(0)}$ denote the baseline corrosion drift estimated sequentially and μ_k^{eff} the corresponding effective corrosion rate after accounting for accumulated damage:

$$\mu_k^{\text{eff}} = \mu_k^{(0)} \left[1 + (\alpha - 1) \cdot \frac{1}{1 + \exp[-r_s(Z_{k-1} - W_0)]}\right], \quad (5)$$

where $Z_{k-1} = X_{k-1} + S_{k-1}$ is the total equivalent damage, W_0 is the midpoint of the sigmoid transition, α is the asymptotic acceleration factor, and r_s (mm^{-1}) is the steepness of the transition. At zero damage the factor equals $1 + (\alpha - 1)/(1 + e^{r_s W_0}) = 1.38$ for the values in Table IV, so $\mu_k^{(0)}$ is a baseline parameter rather than the initial effective drift, and the transition spans about $2\ln 9/r_s = 8.8$ mm between 10% and 90% of the full acceleration. The relationship thus provides a gradual, continuous increase of the corrosion rate with accumulated damage, avoiding the discontinuity associated with a step-function representation.

The reverse dependency is introduced by allowing the effective shock arrival intensity to depend on the current corrosion state:

$$\lambda_k^{\text{eff}} = \lambda_k^{(0)} \exp(\gamma_1 X_{k-1} + \gamma_2 E_k), \quad (6)$$

where $\lambda_k^{(0)}$ denotes the sequentially updated baseline shock intensity, γ_1 (mm^{-1}) sets the sensitivity of the shock intensity to the corrosion depth, and E_k is an optional standardized environmental covariate (e.g., wave height or current speed) with coefficient γ_2. No environmental covariate is used in the experiments reported here ($E_k = 0$), so the effective shock intensity depends on the corrosion depth only; γ_1 is given in Table IV.

Equations (5)–(6) therefore establish a temporally two-way, state-dependent interaction: accumulated shock damage contributes to acceleration of the corrosion process, whereas the corrosion state modifies the effective shock intensity. Both dependencies act with temporal ordering across successive DBN time slices, avoiding a directed cycle within an individual time slice.

Structural failure criterion. The corrosion and equivalent shock-damage channels are combined through a simplified burst-pressure formulation adapted from DNV-RP-F101 [19]:

$$P_{\text{burst}}(t) = \frac{2\, t_{\text{nom}}\, \text{SMYS}}{D - t_{\text{nom}}} \cdot \frac{1 - d(t)/t_{\text{nom}}}{1 - d(t)/(t_{\text{nom}} M)}, \quad (7)$$

where

$$d(t) = X(t) + S(t)$$

is the effective defect depth and

$$M = \sqrt{1 + 0.8\, \frac{L^2}{D\, t_{\text{nom}}}} \quad (8)$$

is the Folias bulging factor. Here, L is the defect length, D is the pipe diameter, and t_{nom} is the nominal wall thickness. Unlike the recommended practice, which uses the specified minimum tensile strength and a length-correction factor with coefficient 0.31, Eqs. (7)–(8) use SMYS and the Folias factor with coefficient 0.8, both of which lower the predicted burst pressure.

Structural failure in the API-DBN is defined by

$$P_{\text{burst}}(t) \le P_{\text{op}}(t), \tag{9}$$

Because P_{burst} decreases monotonically with d, Eq. (9) is equivalent to $Z \ge d_{\text{crit}}$, where $d_{\text{crit}} = 12.07$ mm (63% of the wall) for the case in Table IV; a single large shock that carries Z past d_{crit} therefore causes immediate failure. No separate adaptive degradation threshold is propagated as an inference parameter. Defect geometry is simplified, and additional failure mechanisms such as girth-weld cracking are not explicitly represented in the present model.

B. Adaptive DBN Formulation and Online Bayesian Updating

The physical degradation model defined above is embedded in a time-sliced Dynamic Bayesian Network so that uncertain degradation states and selected model parameters can be updated when new inspection information becomes available. Each DBN slice spans one inspection interval $(t_{k-1}, t_k]$, $k = 1, \ldots, N_I$. Within a slice the state is propagated in sub-steps of length Δt using Eqs. (2)–(6), with the effective rates re-evaluated at every sub-step from the current state, whereas the baseline parameters change only between slices.

Observation model. Inspection measurements are related explicitly to the latent degradation state rather than being incorporated into the state-transition noise. When an inspection reports total equivalent degradation,

$$O_k = X_k + S_k + v_k, \qquad v_k \sim \mathcal{N}(0, \sigma_v^2), \tag{10}$$

where σ_v^2 denotes the inspection-error variance reported in Table IV. If an inspection technology reports corrosion depth alone, the reduced observation model $O_k = X_k + v_k$ can be used. Equation (10) is used in the reported experiments and yields the particle-filter likelihood

$$p(O_k \mid \mathbf{s}_k) = \mathcal{N}\left(O_k; X_k + S_k, \sigma_v^2\right).$$

Temporal DBN formulation. Time-sliced conditional dependencies follow the standard DBN representation [7]. The joint distribution is written as

$$p(\mathbf{s}_{0:N_I}, \theta_{0:N_I}, o_{1:N_I}) = p(\mathbf{s}_0)\, p(\theta_0) \prod_{k=1}^{N_I} p(o_k \mid \mathbf{s}_k)\, p(\mathbf{s}_k \mid \mathbf{s}_{k-1}, \theta_k)\, p(\theta_k \mid \theta_{k-1}), \tag{11}$$

where $p(\mathbf{s}_0)$ and $p(\theta_0)$ are the initial distributions of Algorithm 1, $p(o_k \mid \mathbf{s}_k)$ is the observation likelihood in Eq. (10), $p(\mathbf{s}_k \mid \mathbf{s}_{k-1}, \theta_k)$ represents the physics-informed state transition induced by Eqs. (1)–(4) together with the coupling in Eqs. (5)–(6), and $p(\theta_k \mid \theta_{k-1})$ describes the sequential evolution of the adaptive baseline parameters. Within each slice the parameters are propagated first and the state then moves from $\mathbf{s}_{k-1}$ to $\mathbf{s}_k$ under θ_k, which is the order used in Algorithm 1.

Figure 2 shows two consecutive DBN time slices. Solid arcs represent the state, parameter, and observation dependencies, whereas the dashed arcs denote the temporally ordered corrosion–shock coupling of Eqs. (5)–(6): the total damage

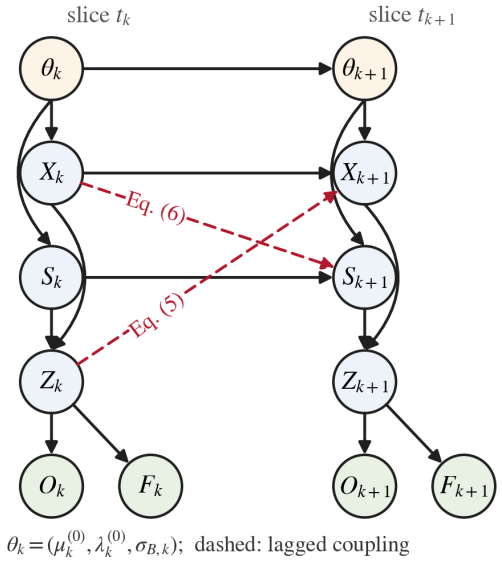


Fig. 2. Two-slice representation of the API-DBN, with $\theta_k = (\mu_k^{(0)}, \lambda_k^{(0)}, \sigma_{B,k})$. Solid arrows denote parameter, state, and observation dependencies; dashed arrows denote the lagged corrosion–shock coupling of Eqs. (5)–(6). All arrows point within a slice or forward in time, so the graph is acyclic.

Z_k affects the corrosion increment of the next slice, and the corrosion depth X_k affects its shock intensity. Every arc points within a slice or forward in time, so the unrolled network is a directed acyclic graph.

Adaptive parameter evolution.

The baseline parameters are constrained to remain positive while being allowed to adapt gradually as additional evidence becomes available. Following the random-walk (artificial-dynamics) approach to parameter tracking in particle methods [17], the baseline parameters evolve between slices as

$$\log \mu_k^{(0)} = \log \mu_{k-1}^{(0)} + \eta_{\mu,k}, \qquad \eta_{\mu,k} \sim \mathcal{N}(0, \sigma_\mu^2), \tag{12}$$

$$\log \lambda_k^{(0)} = \log \lambda_{k-1}^{(0)} + \eta_{\lambda,k}, \qquad \eta_{\lambda,k} \sim \mathcal{N}(0, \sigma_\lambda^2), \tag{13}$$

and

$$\log \sigma_{B,k} = \log \sigma_{B,k-1} + \eta_{\sigma,k}, \qquad \eta_{\sigma,k} \sim \mathcal{N}(0, \sigma_\sigma^2). \tag{14}$$

The effective operating quantities μ_k^{eff} and λ_k^{eff} are subsequently obtained from the baseline parameters through Eqs. (5)–(6). Thus, the stochastic random walks describe gradual evolution of the baseline parameters, whereas the effective parameters represent deterministic state-dependent transformations used in the physical propagation model.

Parameter evolution is assumed Markovian, with no explicit coupling among the random-walk innovations of the different baseline parameters. The standard deviations σ_μ, σ_λ, and σ_σ are specified per year; for a slice of Δ_k years they are multiplied by $\sqrt{\Delta_k}$. Structural failure remains governed solely by Eq. (9); no adaptive failure-threshold parameter is introduced.

Sequential SMC–MCMC inference.

The filtering distribution is approximated using M particles within a bootstrap particle filter augmented by Metropolis–Hastings MCMC rejuvenation. When the effective sample size falls below $M/2$, the particles are resampled systematically, and each resampled particle then receives Metropolis–Hastings moves on its current parameters to reduce particle impoverishment [16].

With the state path held fixed, the move targets $p(\theta_k \mid \theta_{k-1}, \mathbf{s}_{k-1}, \mathbf{s}_k) \propto p(\mathbf{s}_k \mid \mathbf{s}_{k-1}, \theta_k)\, p(\theta_k \mid \theta_{k-1})$, so that a proposal θ^* is accepted with probability

$$\beta = \min\left(1,\ \frac{p(\mathbf{s}_k \mid \mathbf{s}_{k-1}, \theta^*)\, p(\theta^* \mid \theta_{k-1})}{p(\mathbf{s}_k \mid \mathbf{s}_{k-1}, \theta)\, p(\theta \mid \theta_{k-1})}\right). \tag{15}$$

The observation likelihood does not depend on θ_k once the state is fixed and therefore cancels. Because the corrosion increments are Gaussian and the shock counts Poisson given the effective rates, $p(\mathbf{s}_k \mid \mathbf{s}_{k-1}, \theta)$ is available in closed form from each particle's sub-step path. Proposals are Gaussian random walks on $\log \theta_k$ whose standard deviations equal those of the parameter random walk for the slice; five sweeps are applied after each resampling step, and the mean acceptance rate in the reported experiments is 44%. No bounds are imposed after initialization, because the logarithmic parameterization keeps all parameters positive. Algorithm 1 summarizes the complete procedure.

Figure 3 provides the corresponding algorithmic flow. The posterior particle population resulting from the sequential updating stage forms the starting distribution for the RUL prediction and maintenance-decision model developed in Section III-C.

Algorithm 1 Sequential Monte Carlo with MCMC Rejuvenation

Require: Observation likelihood Eq. (10); number of particles M; resampling threshold $M/2$

Initialization: for $m = 1, \ldots, M$, set $\mathbf{s}_0^{(m)} = (0, 0)$ and draw the baseline parameters from their priors: $\mu_0^{(0)} \sim U(0.2, 0.8)$ mm/yr, $\lambda_0^{(0)} \sim \mathrm{Gamma}(\text{shape } 2,\ \text{scale } 0.4)$ truncated to $[0.2, 5]$ yr^{-1}, $\sigma_{B,0} \sim U(0.1, 0.3)$. Set weights $w_0^{(m)} = 1/M$.

for $k = 1$ to N_I **do**

  Parameter evolution: for each m, propagate $\mu_k^{(0),(m)}$, $\lambda_k^{(0),(m)}$, and $\sigma_{B,k}^{(m)}$ according to Eqs. (12)–(14).

  Propagation: for each m and each sub-step of the slice, compute $\mu^{\mathrm{eff},(m)}$ and $\lambda^{\mathrm{eff},(m)}$ from $\theta_k^{(m)}$ and the current state using Eqs. (5)–(6), sample the corrosion increment from Eq. (2) and the shock count from $\mathrm{Poisson}(\lambda^{\mathrm{eff},(m)} \Delta t)$, and accumulate shock damage via Eq. (4).

  if observation o_k available **then**

    Weighting: $w_k^{(m)} \propto w_{k-1}^{(m)} p(o_k \mid x_k^{(m)}, s_k^{(m)})$ via Eq. (10); particles whose path reached d_{crit} during the slice receive zero weight, because the pipeline was found intact; normalize.

    Resampling: compute $\mathrm{ESS} = 1/\sum_m (w_k^{(m)})^2$; if $\mathrm{ESS} < M/2$, resample systematically.

    MCMC rejuvenation: after resampling, apply five Metropolis–Hastings sweeps to each particle's θ_k with acceptance probability β from Eq. (15).

  end if

end for

C. Posterior RUL Prediction and Maintenance Decision Model

The posterior distribution obtained after assimilating the available inspection observations provides a probabilistic representation of the current degradation state and adaptive model parameters. To translate this posterior information into a prognostic quantity, the posterior particle population is propagated forward without further measurements.

Given the filtering distribution at inspection time t_{curr}, $J = 10{,}000$ future degradation trajectories are generated, each starting from a particle drawn from the filtering distribution; along each trajectory the baseline parameters continue their random walk. For trajectory j, the predicted failure time is

$$t_{\mathrm{fail}}^{(j)} = \inf\left\{\tau > t_{\mathrm{curr}} : P_{\mathrm{burst}}(Z(\tau)) \le P_{\mathrm{op}}\right\}, \tag{16}$$

with

$$\mathrm{RUL}^{(j)} = t_{\mathrm{fail}}^{(j)} - t_{\mathrm{curr}}.$$

The resulting ensemble of failure times defines the predictive RUL distribution. The 10th percentile of this distribution is used as a conservative RUL quantity for maintenance decision support.

The maintenance policy is parameterized by the inspection interval I and the preventive-maintenance threshold γ. Inspections take place at $t = I, 2I, \ldots$ within the planning horizon T. At each inspection the filter assimilates the measurement and the RUL forecast is recomputed; preventive replacement is ordered when the 10th percentile of the forecast RUL is at most γ, that is, when $\Pr(\mathrm{RUL} \le \gamma) \ge 0.1$. A failure between inspections (Eq. (9)) triggers corrective replacement at the failure time. Either replacement resets the damage state to zero while the parameter posterior is retained, because the environment is unchanged, and the inspection schedule continues. The expected discounted lifecycle cost is estimated by simulating this closed loop on N_p pipelines:

$$\mathbb{E}[C_{\mathrm{total}}(I,\gamma)] \approx \frac{1}{N_p}\sum_{n=1}^{N_p}\Big[\sum_{t\in\mathcal{T}_{\mathrm{ins}}}\frac{C_{\mathrm{ins}}}{(1+r)^t} + \sum_{t\in\mathcal{T}_{\mathrm{prev}}^{(n)}}\frac{C_{\mathrm{prev}}}{(1+r)^t} + \sum_{t\in\mathcal{T}_{\mathrm{fail}}^{(n)}}\frac{C_{\mathrm{fail}}+C_{\mathrm{env}}}{(1+r)^t}\Big] \tag{17}$$

where $\mathcal{T}_{\mathrm{ins}}$ is the set of inspection times, $\mathcal{T}_{\mathrm{prev}}^{(n)}$ and $\mathcal{T}_{\mathrm{fail}}^{(n)}$ are the times of the preventive and corrective replacements of pipeline n, C_{ins}, C_{prev}, and C_{fail} are the inspection, preventive-repair, and corrective-repair costs, C_{env} is the environmental penalty incurred at failure, and r is the discount rate. Because the costs are those incurred by the simulated pipelines, a model is not rewarded for forecasting longer lives.

The optimal maintenance policy is defined as

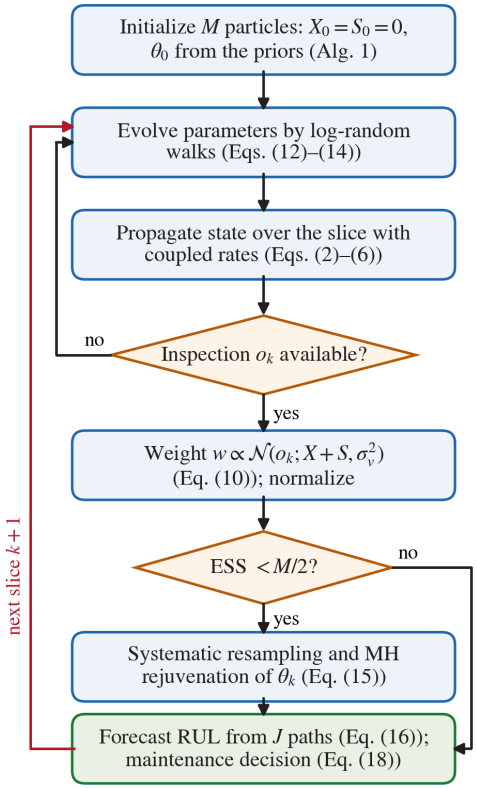


Fig. 3. Particle-filter workflow corresponding to Algorithm 1: parameter evolution, state propagation, observation weighting, resampling, and MCMC rejuvenation, followed by the RUL forecast and maintenance decision of Section III-C.

$$(I^*, \gamma^*) = \arg\min_{I,\gamma} \mathbb{E}[C_{\text{total}}(I, \gamma)]. \tag{18}$$

The search grid, planning horizon, and economic parameters are given in Section IV-C.

IV. Experimental Setup and Evaluation Protocol

The API-DBN is evaluated using two complementary sources of information. Historical BSEE/NETL Gulf of Mexico pipeline and incident records are used to check the parameter assumptions, whereas controlled simulations provide trajectories with known failure times for quantitative assessment of RUL prediction. This distinction is necessary because the historical records do not contain complete longitudinal inspection histories or true RUL labels for all pipeline segments. The experimental protocol therefore labels each parameter as an assumption or a case value, states which of them were checked against the records, and evaluates predictive performance only where ground truth is available.

A. Historical Records

Historical information is taken from the U.S. Offshore Pipeline and Reported Incident Datasets, which NETL compiled from BSEE, PHMSA, and other public sources [2], [3]. Four files of the Energy Data eXchange submission were downloaded on 29 September 2026: the incident table `pipeline_incidents.csv` (uploaded 11 January 2024; 943 incident reports with 85 fields), the location table `pipeline_locations.csv` (uploaded 17 January 2024; 89,986 points along 14,241 BSEE pipeline segments, with 495 columns), and their field dictionaries `pipeline_incidents_data_dictionary.csv` and `pipeline_location_data_dictionary.csv`. Table III summarizes the records.

TABLE III
Summary of the BSEE/NETL Records

Quantity	Value
Incident reports (PHMSA / BSEE)	943 (889 / 54)
Years of the incidents	1986–2021
Reports with lease block (distinct blocks)	881 (487)
Internal / external corrosion flagged	355 / 94
Outside-force damage flagged	158
Diameter / wall / SMYS reported	779 / 734 / 705
Positive reported cost	831
Location points	89,986
Installed segments (with size code)	14,202 (13,792)
Segments with known exposure	11,236
Segment-years, 1986–2021	185,497

The records are processed as follows. The year of an incident is taken from its reported local date. Diameter, wall thickness, and SMYS are treated as missing when empty or non-positive, wall thicknesses above 2 in and SMYS values above 120 ksi are treated as invalid, and SMYS values below 1,000 are read as ksi. Missing values are not imputed: each statistic uses only the records in which the field is reported, and Table III gives the number of such records. Segments are identified by their BSEE segment number; diameter, length, status, and dates are identical on all points of a segment. Proposed and cancelled segments are excluded. The exposure of a segment runs from its installation date, or 1 January 1986 if later, to its abandonment date, or 31 December 2021 if it is later or absent; segments without an installation date, and segments without an abandonment date that are neither active nor out of service, are left out of the exposure. Costs are the reported totals in 2021 U.S. dollars. The records are not partitioned into training, validation, and test sets, because no model parameter is fitted to them and no held-out evaluation is reported (Section V-C).

The records serve three checks of the simulation assumptions, not parameter estimation. First, the case segment of Table IV (outside diameter 0.61 m, or 24 in; wall 19.1 mm, or 0.75 in; API 5L X52) is a large trunkline rather than a typical segment: 1.5% of the installed segments with a size code are 24 in or larger (median 6 in), and the median wall thickness of the pipes involved in incidents is 0.41 in (10.4 mm). Its grade is the most frequently reported one (X52 in 250 of the 705 reports that give SMYS). Second, the 943 reports correspond to 0.0051 incidents per segment-year of exposure. This is an upper bound, because segments with unknown dates are left out of the exposure, and it is far below the modelled shock rate $\lambda = 0.8\ \text{yr}^{-1}$: shocks in the present model include minor impacts that do not produce reportable incidents, so the incident frequency is not an estimate of λ, which remains an

TABLE IV
MODEL PARAMETERS USED IN SIMULATION, WITH SOURCE

Parameter	Value	Description	Source
μ_{base}	0.44 mm/yr	Baseline corrosion drift	Assumption
σ_B	0.20 mm/yr$^{1/2}$	Diffusion coefficient	Assumption
λ	0.8 yr^{-1}	Baseline shock rate	Assumption (Sec. IV-A)
μ_{y_0}	0.32 mm	Mean shock magnitude Y	Assumption
σ_{y_0}	0.15 mm	Std. dev. of Y	Assumption
W_0	1.0 mm	Sigmoid midpoint	Assumption; sensitivity in Sec. IV-A
α	2.0	Acceleration factor	Assumption; sensitivity in Sec. IV-A
r_s	0.5 mm^{-1}	Sigmoid steepness	Assumption; sensitivity in Sec. IV-A
M	1,000	Number of particles	Convergence check, Sec. IV-A
K	200	Time steps (Δt = 0.1 yr, 20-yr horizon)	Assumption
J	10,000	RUL forecast trajectories	Convergence check, Sec. IV-A
κ	0.30	Shock-to-equivalent-damage scaling	Assumption
η	0.50	Shock-damage exponent	Assumption
γ_1	0.05 mm^{-1}	Corrosion effect on shock intensity	Assumption
σ_μ	0.05	Drift random-walk SD (log scale, per year)	Assumption (parameter tracking)
σ_λ	0.05	Shock-rate random-walk SD (log scale, per year)	Assumption (parameter tracking)
σ_σ	0.05	Diffusion random-walk SD (log scale, per year)	Assumption (parameter tracking)
σ_v	0.10 mm	Inspection error SD	Modelling assumption (idealized inspection accuracy)
D	0.61 m	Pipeline diameter	Case value (24-in trunkline; Sec. IV-A)
t_{nom}	19.1 mm	Nominal wall thickness	Case value
L	0.50 m	Defect length	Case value
SMYS	358 MPa	Material yield strength	Case value (API 5L X52; Sec. IV-A)
P_{op}	10 MPa	Operating pressure	Case value

assumption. Third, the failure cost $C_{\text{fail}} = \$5\text{M}$ of Table V exceeds the reported cost of 91.7% of the 831 incidents with a positive cost (median \$0.33M, mean \$2.84M), so it represents a severe failure rather than a typical incident.

Table IV records the value and provenance of each quantity used in the controlled simulations.

Sensitivity and convergence. The sensitivity of the simulated lifetime to the coupling parameters was examined by varying one parameter at a time, with the others at their Table IV values (10,000 trajectories each). The median time to failure changes little for W_0 between 0.25 and 2 mm (12.3–12.9 yr) and for r_s between 0.25 and 5 mm^{-1} (12.2–13.0 yr), but strongly with the acceleration factor: $\alpha = 1$, 2, and 3 give 19.9, 12.5, and 9.2 yr. The value of α is therefore the dominant coupling assumption. Filter accuracy was insensitive to the particle number: on the same 80 pipelines, the RUL MAE of API-DBN varied between 1.01 and 1.04 yr for $M \in \{250, 500, 1000, 2000\}$, so $M = 1{,}000$ is conservative. With $J = 10{,}000$ forecast trajectories, the Monte Carlo standard deviation of the 2.5th and 97.5th RUL percentiles was 0.02 and 0.06 yr (0.07 and 0.15 yr with $J = 1{,}000$).

B. Controlled Simulation Design, Baselines, and Ablations

Quantitative RUL evaluation is performed using controlled degradation simulations because the corresponding failure times are known exactly within the simulated environment. Each simulated pipeline receives its own baseline parameters $(\mu^{(0)}, \lambda^{(0)}, \sigma_B)$, equal to the Table IV values multiplied by independent $\exp(\varepsilon)$ with $\varepsilon \sim \mathcal{N}(0, 0.25^2)$, which then drift annually by the log-random walks of Eqs. (12)–(14) with the standard deviations in Table IV. Degradation follows the corrosion, shock, coupling, and structural-failure formulations of Section III-A, simulated with $\Delta t = 0.1$ yr ($K = 200$ steps over a 20-yr monitoring window) until $Z \geq d_{\text{crit}}$. Each pipeline is inspected annually from $t = 1$ yr; every inspection before failure yields an observation through Eq. (10) with independent Gaussian noise of standard deviation σ_v, and no observations are missing. Performance statistics are evaluated over 30 independent runs with random seeds 0–29, each containing 333 pipelines (9,990 in total). A second design, in which every pipeline has exactly the Table IV parameters, is reported for comparison.

Because the controlled trajectories are generated from the same model family used for API-DBN inference, the experiment evaluates performance under model-consistent conditions; it does not establish robustness to model-form misspecification.

Five model configurations are evaluated under the same controlled simulation conditions:

1) *Fixed Wiener*: a linear Wiener process on the total damage Z whose drift and diffusion are fitted by moments to the inspection increments of 2,000 separately simulated training pipelines; the latest observation is taken as the current state, and the RUL follows the inverse-Gaussian first-passage distribution;
2) *Static DBN*: the proposed model and particle filter with the parameters fixed at the Table IV values, that is, offline calibration without parameter updating, following the static formulation of [8];
3) *PF without MCMC*: the adaptive particle filter without the MCMC rejuvenation step;
4) *No coupling*: the adaptive model with MCMC rejuvenation in which the coupling of Eqs. (5)–(6) is disabled ($\alpha = 1$, $\gamma_1 = 0$); and
5) *API-DBN*: the complete proposed framework with sequential state and parameter updating, MCMC rejuvenation, and state-dependent corrosion–shock coupling.

Two reference configurations complete the analysis: the adaptive model without both coupling and MCMC, which makes the ablation a full 2×2 factorial design, and an oracle that knows each pipeline's true parameters and therefore bounds the achievable error.

The same random seeds and corresponding trajectory sets are used across the compared models. This matched design reduces variation associated with different simulated realizations and supports paired statistical comparison of the resulting performance metrics.

The four adaptive configurations form a 2×2 factorial design in coupling and MCMC rejuvenation, so the effect of each component can be isolated; the static DBN and fixed Wiener models differ from them in more than one respect and serve as external baselines.

C. Evaluation Metrics, Statistical Tests, and Computational Protocol

RUL prediction is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Continuous Ranked Probability Score (CRPS), Brier Score, and 95% Prediction Interval Coverage Probability (PICP). MAE and RMSE quantify point-prediction error, whereas CRPS evaluates the complete predictive RUL distribution. The Brier Score evaluates probabilistic failure prediction within the specified evaluation window, and PICP measures empirical coverage of the nominal 95% RUL prediction interval.

For a predictive RUL distribution F and realized outcome y, CRPS is defined as

$$\mathrm{CRPS}(F, y) = \mathbb{E}_F|X - y| - \tfrac{1}{2}\mathbb{E}_F|X - X'|, \qquad (19)$$

where X and X' are independent draws from F.

The Brier Score is computed as

$$\mathrm{Brier} = \frac{1}{N}\sum_{i=1}^{N}\left(p_i - \mathbb{1}[F_i]\right)^2, \qquad (20)$$

where p_i is the predicted probability of failure for test case i, and $\mathbb{1}[F_i]$ indicates whether failure occurs within the evaluation window.

Prediction-interval coverage is evaluated as

$$\mathrm{PICP} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{1}\left[y_i \in CI_{95\%,i}\right], \qquad (21)$$

where $CI_{95\%,i}$ is the nominal 95% prediction interval for case i. Here, N denotes the number of test cases: one forecast at every annual inspection before failure, giving $N = 121{,}091$ forecasts per model for the 9,990 test pipelines. The $J = 10{,}000$ Monte Carlo trajectories generated for an individual case are used to construct its predictive RUL distribution and associated interval; they are not treated as independent ground-truth observations. The point prediction is the median of the predictive distribution, and the Brier score uses the event that failure occurs within 2 yr of the inspection.

Statistical comparisons use the same 30 matched simulation runs described in Section IV-B. Because each model is applied to the same random-seed realization, the per-run metric values form paired samples. Differences between API-DBN and the comparison models are therefore assessed using a paired t-test [20]. For paired observations with mean difference $\bar{d}$ and standard deviation of the differences s_d, the test statistic is

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \qquad \text{d.f.} = n - 1 = 29, \quad n = 30. \qquad (22)$$

The resulting statistical outcomes are reported with the corresponding predictive-performance results in Section V-A, rather than in the present experimental-protocol section.

Filtering-stage intervals are obtained from the $M = 1{,}000$ particles. The 95% prediction interval of each RUL forecast is the central interval between the 2.5th and 97.5th percentiles of the J simulated failure times [21]. Because every trajectory starts from a particle drawn from the joint state–parameter posterior, the interval reflects state, parameter, and future-process uncertainty.

Maintenance-policy evaluation. The maintenance policy defined by Eq. (18) is evaluated by grid search over $I \in \{1.0, 1.5, \ldots, 4.0\}$ yr and $\gamma \in \{0.5, 1.0, \ldots, 5.0\}$ yr over a planning horizon of $T = 20$ yr. For each candidate policy, Eq. (17) is estimated by running the closed loop of Section III-C on the same $N_p = 1{,}500$ simulated pipelines, generated as in Section IV-B; each pipeline's corrosion noise, shocks, parameter drift, and inspection errors are drawn once and shared by all policies and models (common random numbers). Each replacement decision uses 2,000 forecast trajectories, which estimate the 10% exceedance probability with a standard error of about 0.007.

The economic parameters used in the maintenance experiment are listed in Table V.

Computational protocol. The API-DBN experiments use $M = 1{,}000$ particles and $K = 200$ time steps. Filtering scales as $O(MK)$, MCMC rejuvenation adds a per-particle cost whenever the resampling criterion is satisfied, and each RUL forecast scales as $O(JH)$ for a forecast horizon of H steps. Runtimes were measured on an Apple M4 processor (10 cores, 16 GB RAM) running macOS, with Python 3.14, NumPy 2.5, and SciPy 1.18, one pipeline per core, and averaged over the 30 runs. Observed computational performance is reported in Section V-D.

TABLE V
COST PARAMETERS, WITH SOURCE

Parameter	Description	Value ($k)	Source
C_{ins}	AUV inspection	50	Assumption
C_{prev}	Preventive intervention	200	Assumption
C_{fail}	Corrective repair	5,000	Assumption; exceeds 92% of reported incident costs (Sec. IV-A)
C_{env}	Environmental penalty	10,000	Assumption (regulatory-penalty proxy)
r	Discount rate	0.05	Modelling assumption

V. RESULTS AND DISCUSSION

This section presents the predictive, maintenance-decision, and computational results obtained using the experimental

TABLE VI
RUL PREDICTION PERFORMANCE (MEAN ± STD. OVER 30 RUNS)

Model	MAE (yr)	RMSE (yr)	CRPS (yr)	Brier	PICP (%)
Fixed Wiener	1.37 ± 0.08	1.96 ± 0.14	1.04 ± 0.07	0.031 ± 0.002	75.0
Static DBN	1.38 ± 0.09	2.05 ± 0.16	1.12 ± 0.08	0.032 ± 0.002	59.9
PF w/o MCMC	1.12 ± 0.04	1.73 ± 0.08	0.80 ± 0.03	0.022 ± 0.001	95.2
No coupling	2.29 ± 0.07	3.39 ± 0.10	1.67 ± 0.05	0.027 ± 0.001	77.5
API-DBN	$\mathbf{1.12 \pm 0.04}$	$\mathbf{1.72 \pm 0.08}$	$\mathbf{0.80 \pm 0.03}$	$\mathbf{0.021 \pm 0.001}$	**95.3**
No coupling, w/o MCMC	2.29 ± 0.06	3.38 ± 0.10	1.67 ± 0.05	0.027 ± 0.002	77.3
Oracle (true parameters)	0.70 ± 0.03	1.04 ± 0.06	0.50 ± 0.02	0.018 ± 0.001	95.4

TABLE VII
FACTORIAL ABLATION OF THE ADAPTIVE MODEL (MAE, YR)

	Coupling off	Coupling on
MCMC off	2.29	1.12
MCMC on	2.29	1.12

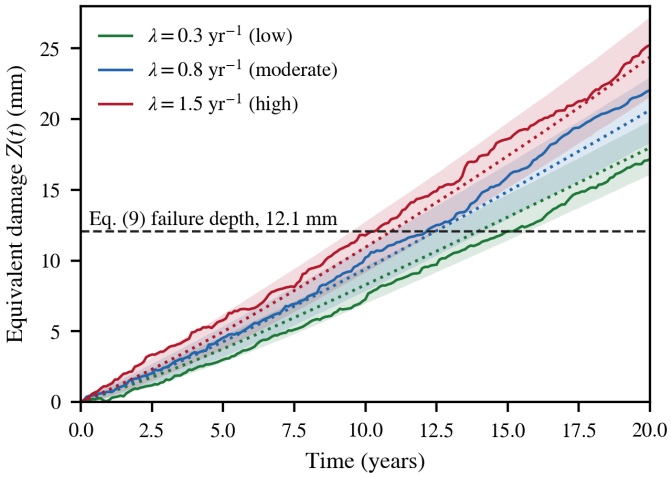


Fig. 4. Equivalent damage simulated with the Table IV parameters for shock rates $\lambda = 0.3$, 0.8, and 1.5 yr^{-1}: one path (solid), the mean (dotted), and the 5–95% band of 10,000 paths (shaded). The dashed line is the depth at which Eq. (9) holds (12.1 mm).

protocol defined in Section IV, and then discusses the implications of adaptive parameter updating, degradation–shock coupling, and posterior uncertainty propagation, together with the limitations of the present formulation.

A. RUL Prediction, Reliability, and Ablation Results

Table VI summarizes RUL prediction over the 30 matched runs (9,990 pipelines; 121,091 forecasts per model). API-DBN attains an MAE of 1.12 ± 0.04 yr against 1.38 ± 0.09 yr for the static DBN and 1.37 ± 0.08 yr for the fixed-parameter Wiener model, a reduction of 18.6%, and it reduces the CRPS by 28.7% (0.80 versus 1.12 yr) and the RMSE by 16%. The larger difference is in calibration: the 95% prediction intervals of API-DBN contain the true RUL in 95.3% of forecasts (Wilson 95% interval 95.2–95.4%), whereas the static DBN, which cannot learn pipeline-specific rates, covers only 59.9% with intervals less than half as wide (2.4 versus 6.1 yr). An oracle that knows each pipeline's true parameters reaches an MAE of 0.70 yr, which bounds what any model can achieve; API-DBN closes 38% of the gap between the static DBN and this bound.

Paired t-tests over the 30 matched runs confirm that API-DBN improves on the static DBN for every metric: MAE $t_{29} = 17.7$, RMSE $t_{29} = 13.8$, CRPS $t_{29} = 23.6$, and Brier score $t_{29} = 29.7$ (all $p < 10^{-13}$). The Brier scores are small for all models because failure within 2 yr of an inspection is a minority event among the pooled forecasts.

Ablation results. Table VII isolates the two components within the adaptive model. Removing the corrosion–shock coupling doubles the MAE, from 1.12 to 2.29 yr: the adaptive drift then has to absorb an acceleration it cannot anticipate, and the RUL is overestimated by 2.1 yr on average. MCMC rejuvenation, in contrast, has no measurable effect with or without coupling (PF without MCMC versus API-DBN: $t_{29} = -0.05$, $p = 0.96$). With the state held fixed, the move of Eq. (15) can only refresh θ_k within the random-walk spread around θ_{k-1}, which the random walk of Eqs. (12)–(14) already provides.

Identical parameters. When every pipeline has exactly the Table IV parameters (3,000 pipelines, 36,090 forecasts per model), the static DBN coincides with the oracle (MAE 0.48 yr) and outperforms API-DBN (0.93 yr; $t_{29} = -56.0$), because the adaptive model must learn parameters that are already known. Sequential parameter updating therefore pays off when parameters differ between assets or drift over time, which is the situation it is designed for.

Degradation trajectories. Figure 4 shows equivalent-damage trajectories simulated with the Table IV parameters for shock rates $\lambda = 0.3$, 0.8, and 1.5 yr^{-1}. Mean damage at 20 yr is 18.0, 20.6, and 24.4 mm, and the median time to reach the Eq. (9) failure depth of 12.1 mm is 14.0, 12.5, and 11.0 yr, respectively. Because the sigmoid factor of Eq. (5) is at least 1.38, the corrosion drift never falls below 0.61 mm/yr, so every simulated pipeline fails within 16 yr even at the lowest shock rate.

Reliability. Figure 5(a) compares the true reliability of the simulated population with the reliability each model implies before any inspection, and Fig. 5(b) shows the conditional reliability of one pipeline after six annual inspections. With the Table IV parameters the true reliability stays above 0.99 until 10 yr and falls below 0.01 by 15 yr (median life 12.5 yr). After six inspections, the API-DBN forecast for the example pipeline lies close to the oracle, whereas the static DBN, which applies the population parameters, predicts a later failure. A reliability curve is a forecast; its quality is judged by the calibration metrics of Section IV-C, not by its height.

B. Maintenance Optimization Results

Each model operated the same 1,500 simulated pipelines over the 20-yr horizon with the closed loop of Section III-C. Figure 6 shows the expected discounted lifecycle cost over the policy grid, and Table VIII lists the cost-minimizing policies. With the static DBN, the expected cost is minimized at $I = 2.5$ yr and $\gamma = 4.5$ yr (\$425k); with API-DBN it is minimized at $I = 4.0$ yr and $\gamma = 5.0$ yr (\$353k), which is 17% lower and 3% above the cost of a policy based on the true parameters (\$342k). The API-DBN and oracle minima lie at the longest inspection interval considered

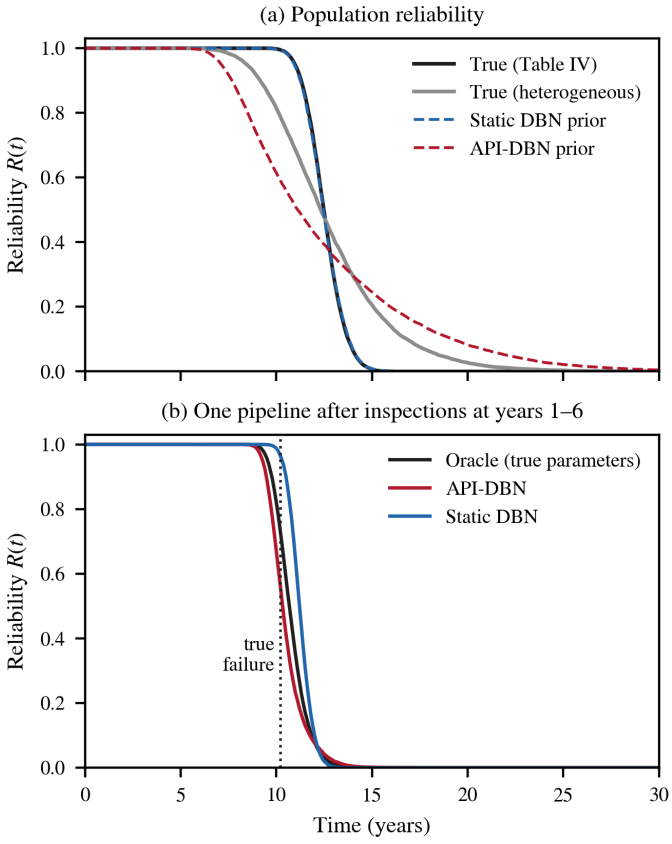


Fig. 5. (a) Reliability of the simulated population with the Table IV parameters and with pipeline-to-pipeline variation, and the reliability implied by the static DBN and by the API-DBN prior before any inspection. (b) Conditional reliability of one pipeline with heterogeneous parameters after inspections at years 1–6; the dotted line marks its true failure time.

TABLE VIII
COST-MINIMIZING MAINTENANCE POLICIES (1,500 PIPELINES, 20-YR HORIZON)

Decision model	I^* (yr)	γ^* (yr)	Cost (\$k)	Pipelines failing (%)
Static DBN	2.5	4.5	425.0 ± 4.2	0.07
API-DBN	4.0	5.0	353.5 ± 4.8	0.07
Oracle (true parameters)	4.0	4.5	342.1 ± 11.1	0.27

($I = 4$ yr), and the API-DBN minimum also lies at the largest threshold ($\gamma = 5$ yr), so policies beyond the searched domain could lower these costs further; the static-DBN minimum lies inside the domain. Policies with $\gamma < I$ leave degraded pipelines unprotected until the next inspection: at $I = 2$ yr and $\gamma = 1.5$ yr, 13–21% of pipelines fail and the expected cost rises to \$1.4–2.3M. The change in prognostic information produced by sequential state and parameter updating therefore propagates to the maintenance-decision layer and lowers the cost of the policy it selects.

C. Use of Historical Records

The BSEE/NETL records are used only for the checks of Section IV-A. They contain neither longitudinal inspection histories nor failure-time labels, so they cannot be used to validate RUL prediction directly. A quantitative evaluation of incident risk, for example the discrimination and calibration of predicted segment-level failure probabilities against incidents reported after a calibration period, is a natural next step. It is not reported here, the records are not partitioned for it, and no claim of historical validation is made.

D. Computational Performance

With $M = 1{,}000$ particles, filtering over the $K = 200$ time steps takes 0.05–0.07 s per pipeline for API-DBN (0.04 s without MCMC rejuvenation), and a complete evaluation including about 12 RUL forecasts with $J = 10{,}000$ trajectories takes 1.1–1.3 s, measured on the platform described in Section IV-C. The forecasts, not the $O(MK)$ filter, therefore dominate the runtime. Because pipelines are independent, the computation parallelizes across cores: all seven configurations were evaluated for the 9,990 pipelines in about 2 h on the 10-core processor.

The measured runtime indicates that the proposed inference procedure is computationally tractable at the scale evaluated in this study. The result should nevertheless be interpreted as an implementation benchmark under the reported experimental configuration rather than as evidence of fleet-scale or real-time deployment.

E. Scientific Discussion and Limitations

The controlled experiments indicate that the advantage of the API-DBN arises from updating the baseline parameters as inspection evidence accumulates, and that it appears when parameters differ between pipelines or drift over time: with pipeline-specific, drifting parameters, API-DBN reduces the MAE by 19% and restores nominal interval coverage, whereas with identical, known parameters the static DBN performs better. The posterior error in the corrosion drift of API-DBN falls from about 21% after the first inspection to about 12% after ten and then stabilizes, because the random walk of Eqs. (12)–(14) gradually discounts old evidence; this limits how closely the adaptive model can approach the oracle.

The ablation results support retaining an explicit dependence between progressive corrosion and discrete shock damage: without it, the adaptive model is the least accurate configuration, because it overestimates the remaining life of pipelines whose corrosion accelerates. The MCMC rejuvenation step, as formulated here, does not improve accuracy; a move that targets the parameter posterior given the whole inspection history [16] may be needed for rejuvenation to contribute.

A second implication concerns uncertainty propagation. The API-DBN does not terminate at a point estimate of the current degradation state. The posterior state and parameter population is propagated to a distribution of future failure times, allowing maintenance decisions to depend on predictive uncertainty rather than on a single deterministic RUL estimate. The maintenance results show that this difference in prognostic information can propagate to the decision layer: the API-DBN-based policy produces a lower expected lifecycle cost than the policy obtained from the static-DBN RUL distribution under the same cost model. This interpretation is conditional on the economic assumptions in Table V; the reported optimum

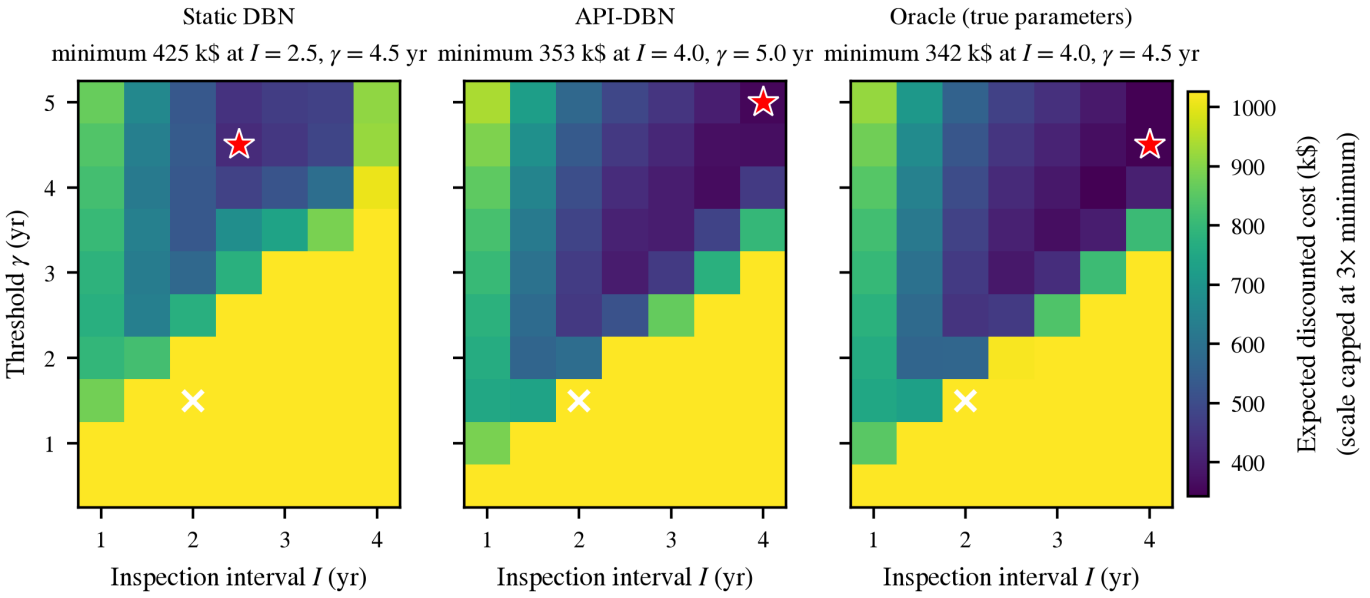


Fig. 6. Expected discounted lifecycle cost (k$) when each model makes the inspection and replacement decisions for the same 1,500 simulated pipelines, over the inspection interval I and the threshold γ on the 10th-percentile RUL. Stars mark the minima; crosses mark the policy $I = 2$ yr, $\gamma = 1.5$ yr. Policies with $\gamma < I$ allow failures between inspections.

should not be assumed to transfer directly to assets with different intervention costs, failure consequences, or operational constraints.

The results should also be interpreted in view of the controlled simulation design. The synthetic trajectories are generated from the same general degradation-process family used by the inference model. They therefore provide a controlled assessment under model-consistent conditions but do not establish robustness to degradation-model misspecification. Evaluation under alternative corrosion laws, shock distributions, or coupling relationships would be required to quantify sensitivity to structural model error.

Several modeling assumptions further limit the present formulation. The Wiener process provides a tractable representation of stochastic corrosion evolution but permits non-monotonic latent increments, whereas physical wall loss is intrinsically cumulative. A Gamma process or another monotonic degradation model may therefore be more appropriate for applications in which this constraint is critical. The equivalent shock-damage mapping in Eq. (4) also reduces heterogeneous mechanical shock mechanisms to an equivalent wall-loss quantity. Its coefficients must be calibrated for the relevant material, geometry, and loading conditions before asset-specific application. The inspection error of 0.1 mm represents an idealized measurement; typical in-line-inspection sizing tolerances are considerably larger, and larger measurement error would slow parameter learning.

The structural-failure representation is likewise simplified. The present model combines corrosion and equivalent shock damage through the burst-pressure criterion of Eq. (9); it does not include an explicit fatigue crack-growth state. Corrosion–fatigue interaction may motivate the broader reliability problem, but fatigue crack propagation is not estimated by the current API-DBN. Defect geometry is represented in simplified form, and other mechanisms such as girth-weld cracking are outside the fitted model.

The historical-data component introduces a separate limitation. BSEE/NETL incident records allow the parameter assumptions to be checked, but they do not constitute longitudinal condition-monitoring histories, and no held-out incident-risk evaluation is reported here. The absence of historical RUL labels prevents direct field-data validation of the RUL errors reported in Table VI; the simulation results therefore support quantitative RUL accuracy under controlled, model-consistent conditions only.

Finally, MCMC rejuvenation roughly doubles the filtering cost relative to a bootstrap particle filter but adds little to the total runtime, which is dominated by the RUL forecasts. Practical deployment would require integration with the available in-line inspection data stream and calibration to the geometry and operating conditions of the target pipeline. Per-segment parallelization provides a route toward larger-scale implementation, but fleet-wide deployment and integration with existing pipeline integrity-management systems have not been demonstrated in the present study.

VI. Conclusion and Future Work

This study developed an Adaptive Physics-Informed Dynamic Bayesian Network (API-DBN) for reliability assessment and maintenance optimization of subsea pipelines subject to interacting progressive corrosion and discrete shock damage. The framework integrates a physics-informed degradation model, temporally ordered corrosion–shock coupling, sequential Bayesian state and parameter updating through SMC–MCMC inference, probabilistic RUL prediction, and risk–

cost-based maintenance optimization within a unified decision-support framework.

The controlled simulation results show that, when degradation parameters differ between pipelines and drift over time, the API-DBN reduces the RUL mean absolute error by 19% and the continuous ranked probability score by 29% relative to a static-parameter DBN, and its 95% prediction intervals attain 95.3% coverage against 59.9% for the static DBN. Propagating the posterior RUL distribution into the maintenance-decision model lowers the expected discounted lifecycle cost by 17% relative to the static-DBN-based policy. When all pipelines share the same known parameters, the static DBN remains preferable. Removing the corrosion–shock coupling doubles the prediction error, whereas the MCMC rejuvenation step has no measurable effect. Direct historical validation of RUL prediction remains unavailable because the historical records do not provide longitudinal inspection and failure-time histories.

The results indicate that sequential evidence assimilation and adaptive parameter updating can materially affect both prognostic estimates and the maintenance decisions derived from them. The state-dependent corrosion–shock formulation further provides a means of representing interaction between progressive degradation and discrete damage events rather than treating the two processes as independent. These findings should nevertheless be interpreted within the assumptions of the present model and experimental design.

Several limitations define the priorities for future work. The Wiener degradation model permits non-monotonic latent increments, and the equivalent shock-damage mapping requires asset-specific empirical calibration. The present structural representation also uses simplified defect geometry and does not include an explicit fatigue crack-growth state. Future studies should therefore examine monotonic degradation models, richer defect and fatigue representations, and improved calibration using longitudinal inspection data. Further work is also needed to evaluate model robustness under degradation-model misspecification and realistic in-line-inspection measurement error, to test rejuvenation moves that use the full inspection history, to carry out a held-out incident-risk evaluation with the BSEE/NETL records, to quantify sensitivity to key economic parameters, and to assess computational performance at larger fleet scales. Integration of multi-modal sensing and digital-twin data may provide an additional basis for improving condition updating and field-level validation as more complete inspection histories become available.

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